

1. Administrative & Study Identification

Full Study Title: A Phase 3 Randomized, Double-blind, Placebo-Controlled Clinical Study to Evaluate the Safety, Tolerability, and Immunogenicity of V116 When Administered Concomitantly with Influenza Vaccine in Adults 50 Years of Age or Older **Short Title/ Acronym:** STRIDE-5 (V116-005) **Protocol Number:** V116-005-01 **NCT Number:** NCT05526716 **Sponsor Name:** Merck Sharp & Dohme LLC (MSD) **Investigational Product:** V116 (CAPVAXIVE™) and Quadrivalent Influenza Vaccine (QIV) **Phase of Development:** Phase 3 **Medical Monitor:** Clinical Director, Merck Sharp & Dohme LLC

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety, tolerability, and immunogenicity of V116 when administered concomitantly with Quadrivalent Influenza Vaccine (QIV) compared with V116 administered sequentially with QIV in adults 50 years of age or older. **Secondary Objectives:** To demonstrate the non-inferiority of immune responses to V116 and QIV at 30 days postvaccination and to assess the overall safety profile, including injection-site and systemic adverse events.

2.2 Background & Rationale Complications from invasive pneumococcal disease can lead to hospitalization, organ damage, and death. Many adult pneumococcal disease cases are caused by serotypes not included in older pneumococcal vaccines. Co-administration of pneumococcal vaccines with seasonal influenza vaccines (like QIV) is a vital public health strategy to improve vaccination coverage and patient compliance. This study addresses the clinical need to confirm that concomitant administration of the 21-valent pneumococcal conjugate vaccine (V116) and QIV is both safe and provides an immune response that is non-inferior to sequential administration.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator / Placebo-controlled (using matching placebo for V116) **Blinding:** Double-blind **Randomization:** Randomized (1:1)

3.2 Planned Enrollment & Duration Target **NS:** 1,080 adults **Number of Sites:** 56 locations **Estimated Study Start:** 23-Sep-2022 **Estimated Primary Completion:** 21-Jun-2023

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 50 years. 2. Willing and able to provide informed consent and comply with study procedures. 3. Generally healthy or presenting with stable chronic medical conditions.

4.2 Exclusion Criteria 1. Prior administration of any pneumococcal vaccine. 2. History of severe allergic reaction, hypersensitivity, or anaphylaxis to any component of the study vaccines. 3. Recent administration of systemic immunosuppressive therapy or immunomodulating biologicals.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single dose of V116 and a single dose of QIV. Participants randomized to the concomitant arm receive both on Day 1. Participants in the sequential arm receive QIV + Placebo on Day 1, followed by V116 on Day 30. **Route of Administration:** Intramuscular (IM) injection. **Duration of Treatment:** Two vaccination visits (Day 1 and Day 30) for each participant.

5.2 Concomitant & Prohibited Therapy Permitted: Standard concomitant medications for stable chronic comorbidities. **Prohibited:** Other investigational vaccines, systemic immunosuppressants, or any non-study pneumococcal/influenza vaccines during the active study period.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening/Baseline	Day 1 Informed Consent, Randomization, First Vaccination (V116+QIV or Placebo+QIV), eDiary setup
Interim 1	Day 30	Second Vaccination (Placebo or V116), Safety Labs, Efficacy Blood Draw (Immunogenicity)	
End of Study	Month 6	Final Safety Follow-up, Exit Interview, SAE Tracking	

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Serotype-specific opsonophagocytic activity (OPA) GMTs for V116 and hemagglutination inhibition (HAI) GMTs for QIV at 30 days postvaccination. **Measurement Method:** Central Laboratory Analysis (validated OPA assays for pneumococcal serotypes and HAI assays for influenza).

7.2 Secondary & Exploratory Endpoints Proportion of participants experiencing injection-site adverse events (AEs) and systemic

AEs. Proportion of participants with serious adverse events (SAEs) throughout the study duration.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Patient electronic diary (eDiary) collected for 14 days post-vaccination to track local and systemic reactions. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness, with active monitoring through Month 6. **Data Monitoring Committee (DMC):** External Independent Data Monitoring Committee (eIDMC) managed by the sponsor.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Per-Protocol (PP) set for primary immunogenicity outcomes; Safety Set for all AE/SAE evaluations. **Sample Size Justification:** $N=1,080$ is adequately powered to test the non-inferiority criteria (the statistical non-inferiority criterion for HAI was set at 0.67). **Handling of Missing Data:** Missing serology data are generally excluded from the Per-Protocol immunogenicity analyses; safety endpoints utilize all available data.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits to ensure protocol adherence. **Audit Trail:** Details on eCRF locking, blinded data review, and data clarification forms via standard sponsor quality audit mechanisms.

11. Study Identifiers & Governance

Primary ID: V116-005 **Registry Number:** NCT05526716 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC **Study Title:** STRIDE-5 **Official Regulatory Title:** A Phase 3 Randomized, Double-blind, Placebo-Controlled Clinical Study to Evaluate the Safety, Tolerability, and Immunogenicity of V116 When Administered Concomitantly with Influenza Vaccine in Adults 50 Years of Age or Older

12. Clinical Framework (Pneumococcal Specific)

Primary Condition: Pneumococcal Pneumonia and Invasive Pneumococcal Disease **Diagnosis Threshold:** N/A (Prophylactic Active Immunization) **Medical Methodology:** Concomitant vs.

Sequential Vaccination administration **Optimization Target:**
Serotype-specific OPA GMTs and HAI GMT non-inferiority

Management Pathway: Standard prophylactic vaccination pathway
Education Protocol (HCP): Site training on double-blind vaccine preparation, administration, and AE reporting
Education Protocol (Patient): Onsite training for eDiary utilization and post-vaccination self-monitoring
Quality Audit Mechanism: eDiary compliance tracking and site adherence monitoring

Total Duration: 6 months follow-up **Onsite Visit Cadence:** Day 1 and Day 30 **Remote Monitoring Frequency:** Follow-up safety calls as needed **Checklist Review Interval:** Day 14 post-vaccination (eDiary review)

Role	Name	Signature	Date	---	---	---	---
Principal Investigator	[Insert Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Insert Name]	_____	[DD-MMM-YYYY]				
Lead Statistician	[Insert Name]	_____	[DD-MMM-YYYY]				